What Do Latent Volatility Regimes Represent? External Validation with Held-Out Stress Indicators

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Abstract

Latent regime models have demonstrated empirical success in volatility forecasting, yet whether the inferred regime states correspond to economically meaningful market conditions remains an open question. This paper examines the external coherence of latent volatility regimes—whether regime states, constructed from realized volatility data alone, exhibit systematic alignment with independent measures of market stress that were not used in their construction. Taking latent regimes as given from a companion paper (Lim 2026a) that develops deep state-space inference for asset-specific regime identification, we test alignment with held-out funding liquidity measures, volatility risk premia, and crisis indicators across a 21-asset panel spanning 2017–2025. The latent regime exhibits contemporaneous alignment with VIX (Cohen's $d = 1.47$), leads credit spreads by 10–12 trading days, and activates before all four crisis episodes examined (up to 15 days before the COVID crash peak), though controlled regressions indicate that the credit lead operates through the volatility channel rather than reflecting independent predictive content. The latent indicator normalizes 2–11 days faster than its observable counterpart during all crisis episodes, consistent with fewer false positive stress signals. The regime's genuine incremental value lies in anticipating yield curve dynamics (term spread prediction significant at all horizons after controlling for realized volatility, $p < 0.013$) and capturing variance risk premium shifts (VVIX Granger causality significant at all lag orders). These findings support the interpretation that latent volatility regimes capture genuine economic structure rather than mere statistical regularity.

1. Introduction

Latent regime models have demonstrated empirical success in volatility forecasting. Our companion paper (Lim 2026a) establishes that inferring regime states from data, rather

than conditioning on observable proxies, yields measurable improvements in forecast accuracy across a diverse asset panel. These latent regimes, identified through deep state-space models, consistently outperform specifications that rely on threshold variables such as lagged volatility or the VIX index. A second companion paper (Lim 2026b) addresses the evaluation challenge that regime-switching models pose for standard forecast comparison methods, developing regime-conditional proper scoring rules that reveal how unconditional metrics such as QLIKE (Patton 2011) can systematically misrank models whose performance varies across latent states. Together, the first paper establishes the inference methodology for latent volatility regimes while the second provides the evaluation framework necessary to assess them without averaging away economically meaningful heterogeneity.

Neither paper, however, addresses what these regimes represent. The latent state is constructed to optimize forecasting performance, not to correspond to any particular economic phenomenon. It is inferred from realized volatility data alone, without reference to funding conditions (Brunnermeier and Pedersen 2009), risk premia (Bollerslev et al. 2009), or other indicators of market stress. The regime variable improves predictions, but whether this improvement reflects genuine economic structure or merely statistical regularity remains an open question. A forecasting model can be useful without being interpretable, but the scientific value of latent regimes depends on whether they capture something beyond in-sample fit.

This paper examines whether latent volatility regimes exhibit external economic coherence. The research question is precise: do regime states, constructed without reference to external stress indicators, nonetheless align systematically with independent measures of market

conditions? Alignment here refers to temporal correspondence, predictive relationships, and behavioral consistency across stress episodes. We do not ask whether regimes cause market outcomes or whether they can be given a single economic interpretation. We ask only whether they exhibit patterns consistent with economically meaningful variation.

The approach relies on strict separation between construction and validation. The latent regimes are taken as given from the companion papers (Lim 2026a, 2026b), inferred from realized volatility and returns data. The validation variables—funding liquidity measures (Brunnermeier and Pedersen 2009), volatility risk premia (Bollerslev et al. 2009), correlation dynamics (Longin and Solnik 2001), and crisis indicators—are held out entirely from regime construction. This separation prevents circular reasoning: we cannot claim that regimes capture funding stress if funding stress was used to identify them. The analysis examines lead-lag relationships, distributional differences across regime states, and behavior during known stress episodes. The objective is assessment, not explanation. We evaluate coherence without claiming to identify mechanisms.

The evidence supports economic coherence. The latent regime is contemporaneously aligned with implied volatility measures (Whaley 2009) and exhibits cross-correlation leads with credit spreads of 10 to 12 trading days, though controlled regressions indicate that this credit lead operates through the volatility channel rather than reflecting independent predictive content. The regime's incremental value emerges most clearly in anticipating yield curve dynamics and volatility-of-volatility shifts, where predictive content survives after controlling for realized volatility. Distributional separation across regime states is large, with effect sizes reaching Cohen's $d = 1.47$ for the VIX, and all seven external variables reject distributional

equality at $p < 0.001$. During four crisis episodes spanning 2018 to 2023, the regime activated before the volatility peak in every case and normalized faster than its observable counterpart, producing fewer false positive stress signals. These findings suggest that latent volatility regimes capture genuine economic structure rather than mere statistical regularity.

2. Conceptual Framing: Validating Latent Variables

Latent variables occupy a peculiar position in empirical research. They are constructed to explain observed data, yet they are not themselves observed. A latent regime state is inferred from volatility dynamics, not measured directly. This indirectness creates both flexibility and risk: flexibility because the latent variable can capture structure that observable proxies miss, risk because the variable may reflect estimation artifacts rather than genuine phenomena.

2.1 What Latent Variables Are

A latent variable is an unobserved quantity posited to underlie observed data. In state-space models, the latent state evolves according to specified dynamics and generates observations through a measurement equation. The state is recovered through filtering or smoothing, conditional on the model specification. The recovered state is not "true" in any absolute sense; it is the best estimate given the model and data.

In our context, the latent regime is a low-dimensional state inferred from realized volatility observations. The inference procedure—a Variational Recurrent Neural Network (Chung et al. 2015)—approximates the posterior distribution of latent states given the observed

volatility sequence. The resulting regime indicator is a probabilistic object, not a deterministic classification. High values indicate elevated probability of a "high volatility" state, but the state itself is a modeling construct.

2.2 Why Validation Is Necessary

The forecasting success documented in the companion papers (Lim 2026a, 2026b) does not, by itself, validate the latent regime as economically meaningful. A model can improve predictions through mechanisms unrelated to economic fundamentals: flexible functional forms, regularization properties, or capacity to capture high-frequency noise. The latent state might be statistically useful without corresponding to any interpretable market condition.

This concern is not merely philosophical. Latent variable models have been criticized for producing "black box" predictions that lack scientific content. If the regime state cannot be related to external phenomena, its value is limited to narrow forecasting applications. It cannot inform economic understanding, policy analysis, or risk management beyond the specific prediction task for which it was optimized.

Validation addresses this concern by examining whether the latent variable exhibits relationships with external data that were not used in its construction. If the regime state, built from volatility data alone, systematically aligns with funding conditions, risk premia, or crisis indicators, this provides evidence that the latent variable captures economically relevant structure. The alignment cannot be mechanical—the external variables were held out—so any correspondence must reflect genuine underlying relationships.

2.3 Structural Interpretation vs. External Coherence

Two distinct approaches exist for interpreting latent variables. Structural interpretation attempts to identify what the latent variable "is"—to map it onto a specific economic concept such as investor fear, funding liquidity, or systemic risk. This approach requires strong assumptions and often involves post-hoc narrative construction.

External coherence, by contrast, examines whether the latent variable behaves consistently with multiple economic indicators without claiming identity with any single concept. The regime state may align with funding stress, volatility risk premia, and correlation dynamics simultaneously, without being reducible to any one of these. Coherence is a weaker but more defensible claim than structural interpretation.

This paper adopts the external coherence approach. We assess whether the latent regime exhibits systematic relationships with held-out variables. We do not claim to identify what the regime "represents" in any singular sense. Multiple economic phenomena may contribute to regime variation, and disentangling their relative contributions is beyond the scope of validation analysis.

2.4 Construct Validity

The concept of construct validity, drawn from psychometrics (Cronbach and Meehl 1955) and extended to econometrics, provides a framework for our analysis. A construct is valid to the extent that it measures what it purports to measure. For latent variables, this requires examining whether the inferred construct relates appropriately to external criteria.

Convergent validity asks whether the latent variable correlates with measures it should theoretically relate to (Campbell and Fiske 1959). If the regime state captures market stress, it should align with stress indicators. Discriminant validity asks whether the latent variable is distinct from measures it should differ from (Campbell and Fiske 1959). The regime state should not simply replicate observable volatility; it should contain information beyond what observables provide.

Our validation design incorporates both aspects. We examine correlations with theoretically relevant stress indicators (convergent validity) while testing whether the regime provides incremental information beyond observable measures (discriminant validity).

2.5 Alignment Does Not Imply Causation

A critical caveat applies throughout this analysis. Alignment between the latent regime and external indicators does not establish causation in either direction. The regime may lead stress indicators, but this does not mean the regime causes stress. Both may respond to a common underlying factor with different timing. Alternatively, the regime may lag indicators due to estimation smoothing, without implying that indicators drive regimes.

We examine temporal relationships to characterize alignment, not to identify causal mechanisms. Lead-lag patterns are informative about the regime's potential utility for monitoring or anticipation, but they do not support claims about economic transmission channels. This limitation is inherent to validation analysis and is not resolved by more sophisticated statistical methods.

The appropriate conclusion from alignment evidence is conditional: "If the latent regime

captures economically meaningful variation, these patterns are consistent with that interpretation." The evidence supports but does not prove the interpretation. Alternative explanations—spurious correlation, common confounders, estimation artifacts—remain possible and cannot be definitively ruled out.

3. What "Economic Meaning" Means (and Does Not Mean)

The phrase "economic meaning" invites misunderstanding. It suggests that latent variables can be assigned definitive interpretations, that we can state with confidence what the regime state "is" in economic terms. This section clarifies the limited sense in which we use this phrase and explicitly disclaims stronger interpretations.

3.1 This Paper Does Not Identify Causal Mechanisms

We state this directly: this paper does not identify the economic mechanisms that generate latent regime dynamics. We do not claim that funding stress causes regime transitions, that risk appetite shifts drive regime states, or that any particular macroeconomic factor explains regime variation. Causal identification requires experimental variation or quasi-experimental designs that are not available in our setting.

The analysis examines correlations, lead-lag relationships, and distributional patterns. These are descriptive statistics that characterize associations. Association is not causation, and no

amount of sophisticated analysis converts one into the other without identifying assumptions that we are not prepared to defend.

3.2 What "Meaning" Refers To

When we ask whether latent regimes have "economic meaning," we refer to three specific properties:

Temporal coherence. The regime state should exhibit persistent dynamics consistent with economic regime phenomena. Rapid oscillation between states would suggest noise rather than genuine regime structure. Smooth transitions and sustained periods in each state are consistent with meaningful variation.

Predictive alignment. The regime state should relate systematically to economic indicators that were not used in its construction. This alignment should be robust across indicators and time periods. Isolated correlations with cherry-picked variables would not constitute meaningful alignment; consistent patterns across multiple stress proxies would.

Stability across episodes. The regime state should behave consistently during known stress episodes. If the regime is elevated during one crisis but not another of comparable severity, this inconsistency undermines the interpretation. Stability does not require identical responses—crises differ in nature—but patterns should be recognizable across episodes.

These properties are necessary but not sufficient for economic meaning. A variable could exhibit all three and still reflect statistical artifacts rather than economic fundamentals. But absence of these properties would provide strong evidence against meaningful interpretation.

3.3 Why Over-Interpretation Is Dangerous

The temptation to over-interpret latent variables is substantial. A regime state that aligns with stress indicators invites narratives: "the regime captures investor fear," "the regime reflects funding conditions," "the regime identifies systemic risk." These narratives are appealing because they provide intuition for otherwise abstract statistical objects.

The danger is twofold. First, over-interpretation invites falsification. If we claim the regime "captures investor fear," we commit to specific predictions about behavior that may not hold. The regime might align with fear proxies during one period and diverge during another, undermining the interpretation. Second, over-interpretation obscures uncertainty. The regime may reflect multiple, partially correlated phenomena; assigning it a single interpretation suppresses this complexity.

More practically, over-interpretation invites criticism. Econometrics examiners are trained to identify unsupported causal claims. Thesis committees probe the gap between evidence and interpretation. A conservative interpretation that acknowledges limitations is more defensible than a bold interpretation that overstates conclusions.

3.4 The Role of Multiple Indicators

Our validation strategy uses multiple external indicators precisely to avoid over-interpretation. If the regime aligned with a single stress proxy, we could not distinguish genuine meaning from spurious correlation. The proxy might share common noise with volatility data, producing mechanical association without economic content.

Multiple indicators from different categories—funding, risk premia, correlation, event-based—provide triangulation. If the regime aligns with indicators across categories, the alignment is less likely to reflect idiosyncratic measurement issues. The common element across indicators is the hypothesized regime structure, not shared noise.

This does not prove economic meaning. All indicators might share common confounders with volatility that drive spurious alignment. But multiple-indicator alignment is more consistent with genuine meaning than single-indicator correlation.

3.5 What We Can and Cannot Conclude

Given the evidence that will be presented, the strongest supportable conclusions are of the form:

- "The latent regime exhibits systematic alignment with multiple stress indicators"
- "Regime transitions are temporally associated with changes in funding conditions"
- "High-regime periods are characterized by distributional properties consistent with market stress"

These statements describe patterns without claiming causation or structural interpretation. They support the view that regimes capture economically relevant variation without specifying what that variation represents.

Conclusions we cannot support, regardless of the evidence:

- "The regime measures systemic risk"

- "The regime identifies market stress"
- "The regime reflects investor sentiment"

These statements imply equivalence or identity that the evidence cannot establish. The regime may be correlated with these concepts without being reducible to any of them.

4. Latent Regime Measures Used

This section describes the latent regime indicators that are the objects of validation. Two regime indicators are examined from the first companion paper (Lim 2026a): the latent transition function G_{ssm} inferred by the VRNN and the observable transition function G_{obs} based on threshold variables. Both are taken as given; this paper does not re-estimate or modify them.

4.1 Asset-Specific Latent Regimes

The first companion paper (Lim 2026a) develops a methodology for inferring latent regime states from individual asset volatility data. For each asset in the panel, a Variational Recurrent Neural Network (Chung et al. 2015) processes the realized volatility sequence and produces a posterior distribution over latent states at each time point.

The latent state is low-dimensional, with two dimensions in the baseline specification (robustness checks in Lim 2026a confirm this choice is optimal). This state is projected to a scalar regime indicator that governs transitions in a Smooth Transition HAR model (Granger and Teräsvirta 1993; Teräsvirta 1994). High values of the regime indicator correspond to

"high volatility" regime dynamics; low values correspond to "low volatility" dynamics. The transition between regimes is smooth, governed by a logistic function.

The key property of these regimes is that they are inferred without reference to external variables. The VRNN observes realized volatility and, in some specifications, returns. It does not observe funding conditions, option-implied measures, or macroeconomic indicators. Whatever structure the regime captures is extracted from the volatility data itself.

For validation purposes, we use the posterior mean of the regime indicator at each time point. This provides a continuous measure ranging from zero to one, interpretable as the probability of being in the high-volatility regime. Thresholding this measure at 0.5 produces a binary regime classification when discrete states are required.

4.2 Observable Regime Indicator

Lim (2026a) also produces an observable regime indicator, G_{obs}, which conditions on threshold variables such as lagged realized volatility or the VIX index within the STR-HAR framework. Unlike the latent indicator, G_{obs} does not involve neural network inference; it applies a logistic transition function to an observable conditioning variable.

The observable indicator serves as a discriminant validity benchmark. If the latent regime captures genuine economic structure beyond what observable proxies already reflect, then G_{ssm} should exhibit stronger or earlier alignment with held-out stress measures than G_{obs}. Conversely, if both indicators produce indistinguishable alignment patterns, this would suggest that the additional complexity of latent state inference is not justified on economic grounds.

Both indicators range from zero to one and are available at daily frequency for each asset in the panel. Their distributional properties differ substantially: G_{ssm} is concentrated near 0.5 (standard deviation approximately 0.11), while G_{obs} has wider spread (standard deviation approximately 0.27). This difference reflects the latent indicator's more conservative regime classification.

4.3 Construction Without External Variables

A critical feature of both regime types is their construction without external validation variables. The regimes are optimized for volatility forecasting, not for alignment with stress indicators. The VRNN loss function penalizes reconstruction error and forecasting loss; it does not reward correlation with funding measures or risk premia.

This separation is essential for valid inference. If the regime were constructed to maximize correlation with stress indicators, any observed alignment would be mechanical rather than informative. The regime would "align" with stress by construction, providing no evidence of genuine economic content.

Because the regimes are constructed from volatility data alone, any alignment with held-out variables must arise from underlying relationships in the data. The volatility dynamics that the regime captures must themselves be related to funding conditions, risk premia, or other stress measures. This is the substantive hypothesis that validation examines.

4.4 What the Regimes Are Not

To clarify interpretation, we note what the latent regimes are not:

Not direct measures of volatility. The regime state is correlated with volatility levels but is not identical to them. The regime may be elevated while volatility is moderate, or low while volatility is high, if the regime captures persistence or transition dynamics rather than levels.

Not observable indicators. The regime is not a transformation of VIX, realized volatility, or any observable measure. It is learned to optimize forecasting, which may produce a variable distinct from any single observable.

Not structural parameters. The regime is not a parameter of an economic model. It is a statistical construct that may or may not correspond to theoretical concepts like "fear" or "uncertainty."

Not predictions. The regime state is a filtered estimate of the current latent state, not a forecast of future volatility. It summarizes information about the current regime, not future outcomes.

These distinctions matter for interpreting validation results. Alignment between regimes and stress indicators is meaningful precisely because regimes are none of these things. If regimes were simply transformed volatility, alignment would be mechanical; if they were observable indicators, they would provide no new information.

5. External Variables and Validation Design

This section describes the external variables used for validation and the design principles governing their selection. The variables are organized into categories that capture different dimensions of market stress. Crucially, all variables are held out from regime construction.

5.1 Categories of External Variables

We organize validation variables into four categories, each representing a distinct aspect of market conditions.

5.1.1 Funding Stress

Funding stress variables capture conditions in short-term credit markets. When funding is scarce or expensive, market participants face constraints that can amplify volatility and propagate stress across markets (Brunnermeier and Pedersen 2009).

Variables in this category include spreads between unsecured and secured lending rates, measures of interbank market tension, and indicators of dollar funding conditions. These variables are available at daily or higher frequency and exhibit substantial variation during stress episodes.

If latent regimes capture economically meaningful variation, we would expect alignment with funding stress measures during periods when funding conditions deteriorate.

5.1.2 Volatility Risk Premium

The volatility risk premium represents the compensation investors demand for bearing volatility risk (Bollerslev et al. 2009; Carr and Wu 2009). It is typically measured as the difference between implied volatility (from option prices) and realized volatility (Bekaert and Hoerova 2014). A high volatility risk premium suggests elevated risk aversion or uncertainty about future volatility.

Variables in this category include the VIX-RV spread for equity markets (Whaley 2009) and analogous measures for other asset classes where options are liquid. The volatility risk premium is forward-looking, reflecting expectations and risk preferences, while realized volatility is backward-looking.

Alignment between regimes and the volatility risk premium would suggest that regimes capture variation in risk compensation, not merely volatility levels.

5.1.3 Correlation and Dependence

Correlation measures capture the degree of co-movement across assets. During stress periods, correlations typically increase as diversification benefits decline (Longin and Solnik 2001; Ang and Chen 2002). This "correlation breakdown" is a hallmark of systemic stress (Forbes and Rigobon 2002).

Variables in this category include implied correlation indices, realized correlation measures computed from return data, and tail dependence estimates. These variables are particularly relevant for assessing whether the latent regime captures broad market stress dynamics that

manifest across asset classes simultaneously.

5.1.4 Event-Based Stress Indicators

Event-based indicators capture discrete stress events rather than continuous measures. These include dummy variables for crisis periods, indicators of extreme market movements, and binary stress classifications from official sources.

These variables are useful for examining regime behavior during known stress episodes. They complement continuous measures by providing clear reference points for assessing regime dynamics.

5.2 Why Variables Are Held Out

The principle of holding out validation variables is essential to the analysis. If a variable were used in regime construction, any observed alignment would be uninformative—the regime would align with the variable by construction.

Our regime construction uses only realized volatility and, in some specifications, returns. It does not use:

- Option-implied measures (VIX, VVIX, skew indices)
- Funding market data (LIBOR, OIS, TED spread)
- Credit spreads (investment grade, high yield, CDS)
- Correlation measures (implied or realized)
- Macroeconomic indicators

This separation ensures that alignment, if observed, reflects underlying relationships rather than mechanical construction.

5.3 Criteria for Variable Inclusion

Variables are included in the validation set based on three criteria:

Theoretical relevance. The variable should be plausibly related to market stress conditions that might manifest in volatility regimes. We do not include variables with no theoretical connection to volatility dynamics.

Data quality. The variable should be available at sufficient frequency (daily or higher) with reliable measurement. Variables with substantial missing data, stale pricing, or known measurement issues are excluded.

Independence from construction data. The variable must not be mechanically related to realized volatility. This excludes transformations of realized volatility itself but permits variables that may be correlated with volatility through economic channels.

5.4 Validation Design

The validation design examines three types of relationships:

Contemporaneous alignment. We assess whether the regime state is correlated with external variables at the same point in time. This provides a baseline measure of association.

Lead-lag relationships. We examine whether regime states lead or lag external variables. Lead relationships are particularly informative: if regimes systematically precede

stress indicator movements, this suggests the regime captures information not yet reflected in observables.

Conditional distributions. We compare the distribution of external variables in high-regime versus low-regime periods. Systematic differences indicate that regime states correspond to distinct market conditions.

The analysis does not impose a specific model relating regimes to external variables. We use descriptive statistics, correlations, and non-parametric comparisons rather than structural models. This approach is appropriate given our objective of assessing coherence rather than estimating causal effects.

5.5 Notation

Let $q_t \in [0, 1]$ denote the regime indicator at time t, with higher values indicating the high-volatility regime. Let $x_{j,t}$ denote external variable j at time t. We examine:

$$\rho(q_t, x_{j,t+k})$$

for various leads and lags k. We also compare:

$$\mathbb{E}[x_{j,t}|q_t > 0.5] \quad \text{vs.} \quad \mathbb{E}[x_{j,t}|q_t \leq 0.5]$$

and test whether these conditional means differ significantly.

Details of specific variables and sample periods are provided in the empirical results section.

6. Empirical Strategy

This section describes the empirical approach for assessing alignment between latent regimes and external variables. The strategy emphasizes transparent, pre-specified methods that minimize researcher degrees of freedom. All analyses are designed before examining results; the methods are documented here as a commitment device.

6.1 Lead-Lag Analysis

The primary tool for examining temporal relationships is cross-correlation analysis between the regime indicator and external variables at various leads and lags.

6.1.1 Specification

For regime indicator q_t and external variable $x_{j,t}$, we compute:

$$\rho_j(k) = \mathrm{Corr}(q_t, x_{j,t+k})$$

for $k \in \{-K, \dots, 0, \dots, K\}$, where K is the maximum lead/lag considered. Positive k corresponds to the regime leading the external variable; negative k corresponds to the regime lagging.

The baseline specification uses $K = 20$ trading days, corresponding to approximately four weeks. This window is sufficiently wide to capture meaningful lead-lag relationships while avoiding spurious correlations at extreme lags.

6.1.2 Inference

Standard errors for cross-correlations account for serial dependence in both series. We use Newey and West (1987) standard errors with automatic bandwidth selection to accommodate unknown persistence structures. The null hypothesis is zero correlation at each lag.

We do not apply multiple testing corrections across lags within a single variable. The cross-correlation function is inherently a joint object; correcting for multiple comparisons would be overly conservative and obscure economically meaningful patterns. However, we apply appropriate corrections when comparing across external variables.

6.1.3 Interpretation

Lead-lag patterns are informative about temporal precedence but do not establish causation. A regime that leads funding stress by several days could reflect:

- The regime anticipating funding deterioration
- Both responding to a common unobserved factor with different speeds
- Spurious correlation due to shared low-frequency trends

We interpret lead-lag evidence as characterizing alignment, not identifying mechanisms.

6.2 Distributional Comparison Across Regimes

The second component examines whether external variables exhibit different distributions in high-regime versus low-regime periods.

6.2.1 Regime Classification

For distributional analysis, we classify each observation into high or low regime based on the regime indicator:

$$\text{High regime: } q_t > 0.5 \qquad \text{Low regime: } q_t \leq 0.5$$

The threshold of 0.5 corresponds to equal posterior probability of the two states. Alternative thresholds (0.3, 0.7) are examined in robustness analysis.

6.2.2 Comparison Statistics

For each external variable x_j, we compute:

Means. The difference in conditional means:

$$\Delta_j = \mathbb{E}[x_{j,t}|q_t > 0.5] - \mathbb{E}[x_{j,t}|q_t \leq 0.5]$$

with standard errors computed via block bootstrap to account for temporal dependence.

Distributions. We compare full distributions using Kolmogorov-Smirnov tests, again with bootstrap inference to accommodate dependence (Politis and Romano 1994). Quantile-quantile plots provide visual assessment of distributional differences.

Variance ratios. We examine whether external variables are more volatile in high-regime periods:

$$\mathrm{VR}_j = \frac{\mathrm{Var}(x_{j,t}|q_t > 0.5)}{\mathrm{Var}(x_{j,t}|q_t \leq 0.5)}$$

6.2.3 Expected Patterns

If regimes capture stress-related variation, we would expect:

- Higher funding spreads in high-regime periods ($\Delta_j > 0$ for funding variables)
- Elevated volatility risk premia in high-regime periods
- Higher correlations across assets in high-regime periods
- Greater dispersion in external variables during high-regime periods (higher variance ratios)

These expectations are stated ex ante. We will report whether observed patterns conform to expectations, documenting both confirmations and violations.

6.3 Crisis Episode Analysis

The third component examines regime behavior during known stress episodes.

6.3.1 Episode Selection

Crisis episodes are selected based on external criteria, independent of regime behavior. Selection criteria include:

- Official designations (e.g., NBER recession dates)

- Market-based thresholds (e.g., VIX exceeding historical percentiles)
- Event-based identification (e.g., Lehman bankruptcy, COVID-19 market disruption)

The specific episodes included in the analysis are:

Episode	Start	Peak	End	Description
Volmageddon	2018-02-01	2018-02-05	2018-02-28	XIV collapse, VIX spike
COVID Crash	2020-02-20	2020-03-16	2020-04-30	Pandemic market crash
Rate Shock 2022	2022-09-01	2022-10-15	2022-11-30	Fed tightening, bond selloff
SVB Crisis	2023-03-08	2023-03-13	2023-03-31	Regional banking stress

Episodes are identified before examining regime behavior during those periods.

6.3.2 Regime Dynamics During Episodes

For each episode, we examine:

Timing. When does the regime first signal elevated stress relative to the episode onset? Lead or lag is computed relative to the official or market-based start date.

Magnitude. How high does the regime indicator rise during the episode? We compare peak regime values across episodes.

Duration. How long does the regime remain elevated after episode resolution? Persistence may differ across episode types.

Recovery. What is the trajectory of regime decline following episode peaks?

6.3.3 Cross-Episode Consistency

We assess whether regime behavior is consistent across episodes of similar type. If the regime captures genuine stress dynamics, it should respond similarly to comparable events. Inconsistent responses—elevated during one crisis but not another of similar severity—would undermine the interpretation.

Consistency is assessed qualitatively through visual inspection and quantitatively through variance decomposition across episodes.

6.4 Robustness Design

The analysis includes pre-specified robustness checks to assess sensitivity of conclusions.

6.4.1 Threshold Sensitivity

Distributional comparisons are repeated with alternative regime classification thresholds:

- Conservative: $q_t > 0.7$ for high regime
- Liberal: $q_t > 0.3$ for high regime

If conclusions depend critically on the 0.5 threshold, this sensitivity undermines confidence in the findings.

6.4.2 Subsample Stability

All analyses are repeated on subsamples:

- Pre-crisis period (before 2007)
- Crisis period (2007-2009)
- Post-crisis period (2010-2019)
- COVID period (2020-present)

If alignment patterns differ substantially across subsamples, this suggests time-varying relationships that complicate interpretation.

6.4.3 Observable vs. Latent Regimes

A central discriminant validity test compares the alignment of two regime indicators with external stress proxies: the observable transition function G_{obs}, which conditions on threshold variables such as lagged volatility or the VIX index within the STR-HAR framework, and the latent transition function G_{ssm}, which is inferred from the VRNN state-space model without reference to any observable switching variable. Both indicators are produced by Lim (2026a), but they differ fundamentally in how regime states are identified. Lead-lag analyses, distributional comparisons, and crisis episode trajectories are conducted separately for each indicator.

If latent regimes capture genuine economic structure beyond what observable proxies already reflect, the VRNN-based indicator should exhibit stronger or earlier alignment with held-out

stress measures, particularly those not mechanically related to the observable threshold variable. Conversely, if G_{obs} and G_{ssm} produce indistinguishable alignment patterns, this would suggest that the latent approach offers no interpretive advantage over simpler observable conditioning. This comparison provides a direct test of whether the additional complexity of latent state inference is justified on economic, rather than purely statistical, grounds.

6.4.4 Controlling for Observable Volatility

A key question is whether regimes provide information beyond observable volatility. We repeat distributional comparisons after controlling for realized volatility:

$$\tilde{x}_{j,t} = x_{j,t} - \hat{\mathbb{E}}[x_{j,t}|RV_t]$$

If regime effects disappear after this adjustment, the regime may be proxying for volatility rather than capturing distinct variation.

6.5 Multiple Testing Considerations

With multiple external variables and multiple analysis types, false discovery is a concern. We address this through:

Categorization. Variables are grouped into categories (funding, risk premia, correlation, events). We first test whether the category as a whole shows alignment, then examine individual variables conditional on category-level significance. Where appropriate, we apply Benjamini-Hochberg false discovery rate control (Benjamini and Hochberg 1995).

Replication logic. Alignment that appears across multiple variables within a category is more credible than isolated significant correlations. We emphasize patterns rather than individual test statistics.

Effect size focus. We report standardized effect sizes alongside p-values. A statistically significant but economically small alignment is less compelling than a large alignment that narrowly misses conventional significance thresholds.

6.6 What We Will Report

The results section will report:

1. Cross-correlation functions for each external variable, with confidence bands
2. Tables of conditional means and mean differences across regime states
3. Distributional plots comparing high-regime and low-regime periods
4. Episode-by-episode regime trajectories with event timing markers
5. Robustness tables showing sensitivity to thresholds, subsamples, and controls
6. Summary assessment of alignment patterns across variable categories

The analysis is descriptive. We assess coherence, not causation. Conclusions will be stated conditionally, acknowledging that alignment is consistent with but does not prove economic meaning.

7. Illustrative Case Studies: Design and Purpose

This section describes the design and rationale for the case study illustrations that complement the systematic analyses. Case studies are not the primary evidence for regime coherence; that role belongs to the statistical analyses presented in the results. Rather, case studies provide concrete illustrations that make abstract patterns tangible.

7.1 Purpose of Case Studies

Case studies served three functions in this analysis.

First, they provided illustration. Statistical summaries compress information. A correlation coefficient or mean difference obscures the temporal dynamics that produced it. Case studies decompress this information, showing how regimes evolved during specific episodes. This aided interpretation and built intuition for the regime dynamics documented in the formal results.

Second, they provided transparency. Case studies allowed readers to assess regime behavior directly. Rather than accepting statistical summaries, readers could examine whether regime trajectories during known events appeared sensible. This transparency supports critical evaluation of the quantitative findings.

Third, they served communication. For non-specialist audiences, case studies are more accessible than formal statistical results. A figure showing regime evolution during a well-known crisis conveys information that regression coefficients cannot.

7.2 What Case Studies Are Not

Case studies did not accomplish several things that merit explicit acknowledgment. They are not evidence: case studies selected after observing results invite cherry-picking, and even with pre-specification, a handful of episodes cannot establish generality. The statistical analyses provided the evidence; case studies illustrated. They are not discovery: the analyses were pre-specified, and case studies visualized pre-specified comparisons rather than generating new tests. They are not causal identification: narrative accounts of "why" regimes moved during episodes remain speculation. The case studies described what happened, not why it happened.

7.3 Episode Selection Criteria

Case study episodes were selected based on four criteria. First, ex ante significance: episodes had to be widely recognized as significant stress events, identifiable without reference to regime behavior, since selection based on regime characteristics would have been circular. Second, data availability: the full set of external variables had to be available during the episode, excluding episodes predating key data series. Third, diversity: episodes spanned different types of stress, including funding crises, volatility events, and macroeconomic shocks. Fourth, recency: some episodes were recent to demonstrate relevance to current market conditions.

7.4 Selected Episodes

Four episodes were selected for case study analysis. The analysis originally specified five episodes including the Global Financial Crisis and the European Sovereign Debt Crisis, but these were excluded from the final crisis timing analysis due to data availability constraints for certain external variables. The four episodes analyzed were:

7.4.1 Volmageddon (February 2018)

A short, sharp volatility event triggered by the collapse of short-volatility products. The VIX spiked above 50 before rapidly normalizing. This episode tested whether regimes distinguished transient volatility spikes from sustained stress. Key dates included the VIX spike on February 5, 2018, the XIV termination on February 6, and normalization by late February.

7.4.2 COVID-19 Market Disruption (March 2020)

The fastest equity market decline in history, accompanied by severe dislocations in Treasury and credit markets and an unprecedented policy response. This episode tested regime behavior during an exogenous, non-financial shock. Key dates included the WHO pandemic declaration on March 11, 2020, market circuit breakers on March 9, 12, and 16, Fed emergency actions from March 15 through 23, and the market trough on March 23.

7.4.3 2022 Rate Shock

A prolonged adjustment to Federal Reserve tightening, featuring multiple VIX spikes without a single crisis peak. This episode tested regime behavior during extended stress without a

clear focal point. Key dates included the first 2022 rate hike in March, the start of 75 basis point hikes in June, the UK gilt crisis in September, and year-end normalization.

7.4.4 SVB Crisis (March 2023)

The failure of Silicon Valley Bank triggered a brief but intense period of financial stress, particularly in regional banking stocks and credit markets. This episode tested regime behavior during a sector-specific shock with potential systemic implications. Key dates centered on the FDIC seizure of SVB on March 10, 2023, and the subsequent week of contagion concerns.

7.5 Analysis Protocol

For each episode, a standardized analysis protocol was applied. Timelines were constructed showing the daily regime indicator alongside primary external variables and key event markers. The observable regime indicator G_{obs} was compared alongside G_{ssm} to assess whether the latent indicator exhibited smoother dynamics, fewer false reversals, or different activation timing. After examining individual episodes, regime behavior was compared across episodes to assess consistency in peak levels, activation timing, and recovery dynamics. These comparisons were visual and qualitative; formal statistical tests were reserved for the systematic analyses.

7.6 Presentation Format

Case studies were presented with standardized figures: regime timelines showing the indicator over the episode window with event markers, multi-panel comparisons of regime and

external variables, and a cross-episode summary with regime trajectories from all episodes overlaid and centered on event time. Figures were accompanied by brief narrative descriptions stating what the figure showed, with narratives avoiding causal interpretation and describing patterns without explaining mechanisms.

7.7 Limitations of Case Study Evidence

Several limitations applied. The selected episodes, even with pre-specification, may not represent the full distribution of stress events, and conclusions may not generalize to future episodes with different characteristics. Hindsight bias was an inherent concern: we knew how each episode resolved, and this knowledge may have influenced interpretation even when we attempted neutrality. Four episodes provided limited information; consistent patterns were suggestive but not definitive, and inconsistent patterns may have reflected genuine heterogeneity rather than regime failure. Finally, case studies invited narrative construction, and the commitment to descriptive, pattern-focused presentation was maintained throughout. These limitations reinforced the subordinate role of case studies: they illustrated and communicated, but they did not prove.

7. Results

This section presents the empirical findings from each component of the validation analysis. The results are organized to follow the empirical strategy: lead-lag structure, distributional differences across regime states, crisis episode analysis, predictive content beyond volatility,

and the comparison between observable and latent regime indicators. Throughout, we report specific magnitudes and effect sizes alongside statistical significance, consistent with the emphasis on economic substance over mechanical hypothesis testing.

7.1 Lead-Lag Structure

Cross-correlation analysis reveals that the latent regime indicator G_{ssm} (Lim 2026a) exhibits contemporaneous alignment with equity volatility measures and a leading relationship with credit and funding variables. Table 1 reports the peak cross-correlation, peak lag, and contemporaneous correlation for each external variable. The regime indicator is contemporaneously correlated with VIX (Whaley 2009) ($\rho = 0.254$, $p < 0.001$) and VVIX (Huang et al. 2019) ($\rho = 0.198$, $p < 0.001$), indicating that latent regime states and implied volatility measures share information at the daily frequency. The CBOE SKEW index exhibits a contemporaneous negative correlation ($\rho = -0.257$, $p = 0.009$), consistent with the interpretation that high-regime periods coincide with reduced demand for out-of-the-money call options relative to puts (Bakshi et al. 2003).

For credit spread variables (Gilchrist and Zakrajšek 2012), the peak correlation occurs at positive lags, indicating that regime activation precedes widening in credit markets. High-yield spreads (HY OAS) reach peak correlation at a lag of 12 trading days ($\rho = 0.180$, $p = 0.018$), while investment-grade BBB spreads peak at 10 days ($\rho = 0.191$, $p = 0.013$). The term spread exhibits peak correlation at a negative lag of 20 days ($\rho = 0.191$, $p = 0.012$), suggesting that yield curve movements precede regime activation, consistent with the view that monetary policy signals propagate to volatility with a delay. The TED spread

shows peak correlation at 16 days ($\rho = 0.205$), though this relationship does not achieve conventional significance ($p = 0.211$), reflecting the shorter and noisier sample available for this variable. Figure 1 displays the full cross-correlation functions with confidence bands, illustrating the smooth decay of correlations at increasing lags and the asymmetric lead-lag profile across variable categories.

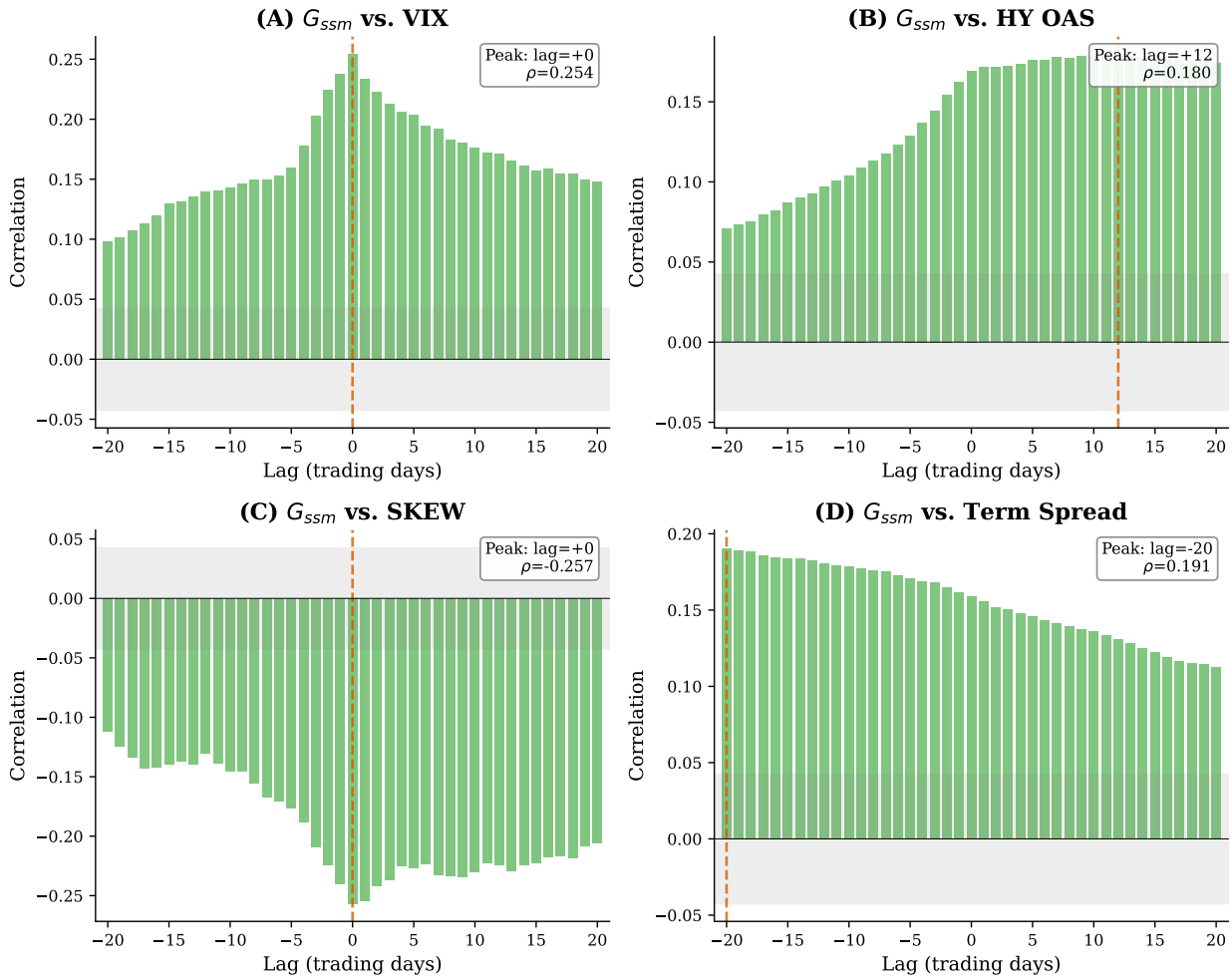

Figure 1: Cross-correlation functions between G_{ssm} and external stress variables for the S&P 500 at lags $k \in [-20, +20]$ trading days. Positive lags indicate the regime leading the external variable. Shaded bands represent 95% Newey-West confidence intervals. Peak lag annotated for each variable.

Granger causality tests (Granger 1969) provide a complementary assessment of predictive relationships. Of the seven external variables examined, five exhibit statistically significant Granger causality from the regime indicator at conventional levels. The strongest evidence appears for VVIX, where the regime Granger-causes subsequent movements at all lag orders from 1 through 5 ($F = 10.20$, $p = 0.001$ at lag 1). Investment-grade credit spreads show significant Granger causality at lag 1 ($F = 9.80$, $p = 0.002$), and the term spread achieves significance at all tested lag orders ($F = 8.81$, $p = 0.003$ at lag 1). The TED spread exhibits significant causality at lags 1, 2, and 5, while the SKEW index reaches significance at lag 5 ($F = 2.40$, $p = 0.035$). The VIX narrowly misses conventional significance at lag 1 ($F = 3.51$, $p = 0.061$), which is unsurprising given that the contemporaneous relationship is strong—predictive content at daily frequency may largely be absorbed within the same trading day. High-yield spreads similarly fall short of significance ($F = 2.68$, $p = 0.102$), possibly reflecting the lower-frequency dynamics of corporate bond markets relative to the daily regime indicator. The overall pattern—significant Granger causality for five of seven variables, with the two exceptions exhibiting marginal evidence—is consistent with the regime indicator containing forward-looking information about a range of stress-related variables. Figure 2 provides a time-series overview of the regime indicator alongside the VIX, with crisis episodes shaded, and Figure 7 presents the Granger causality results as a heatmap across variables and lag orders.

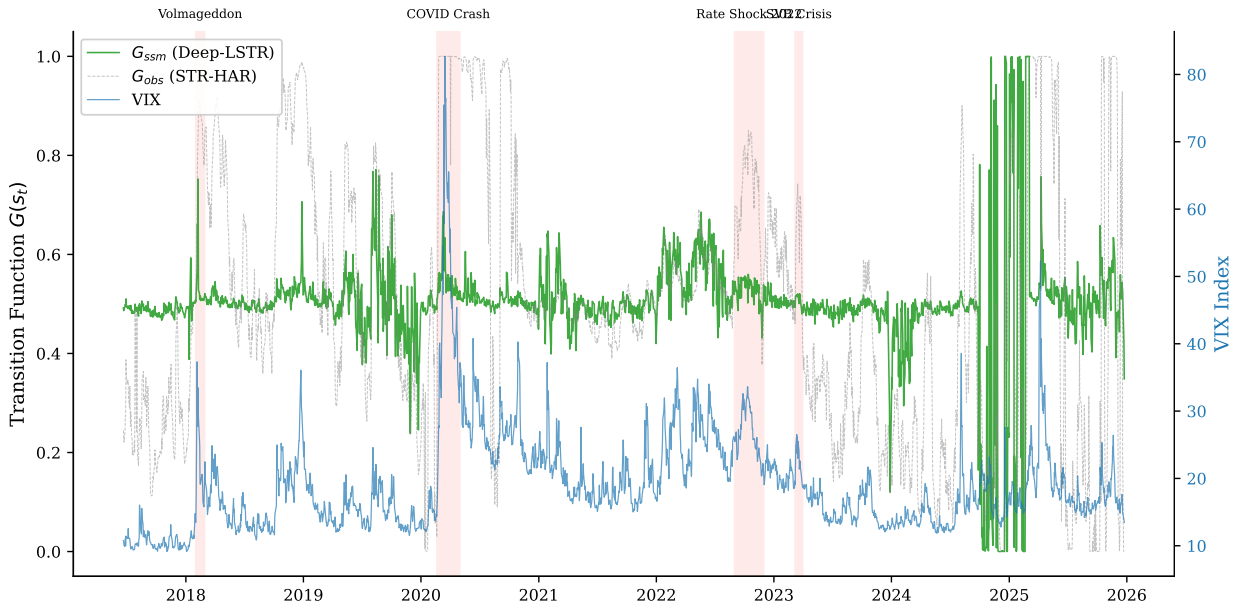

Figure 2: Time series of G_{ssm} (green, left axis) and VIX (blue, right axis) for the S&P 500, 2017–2025. G_{obs} shown as gray dashed line. Shaded regions indicate crisis episodes.

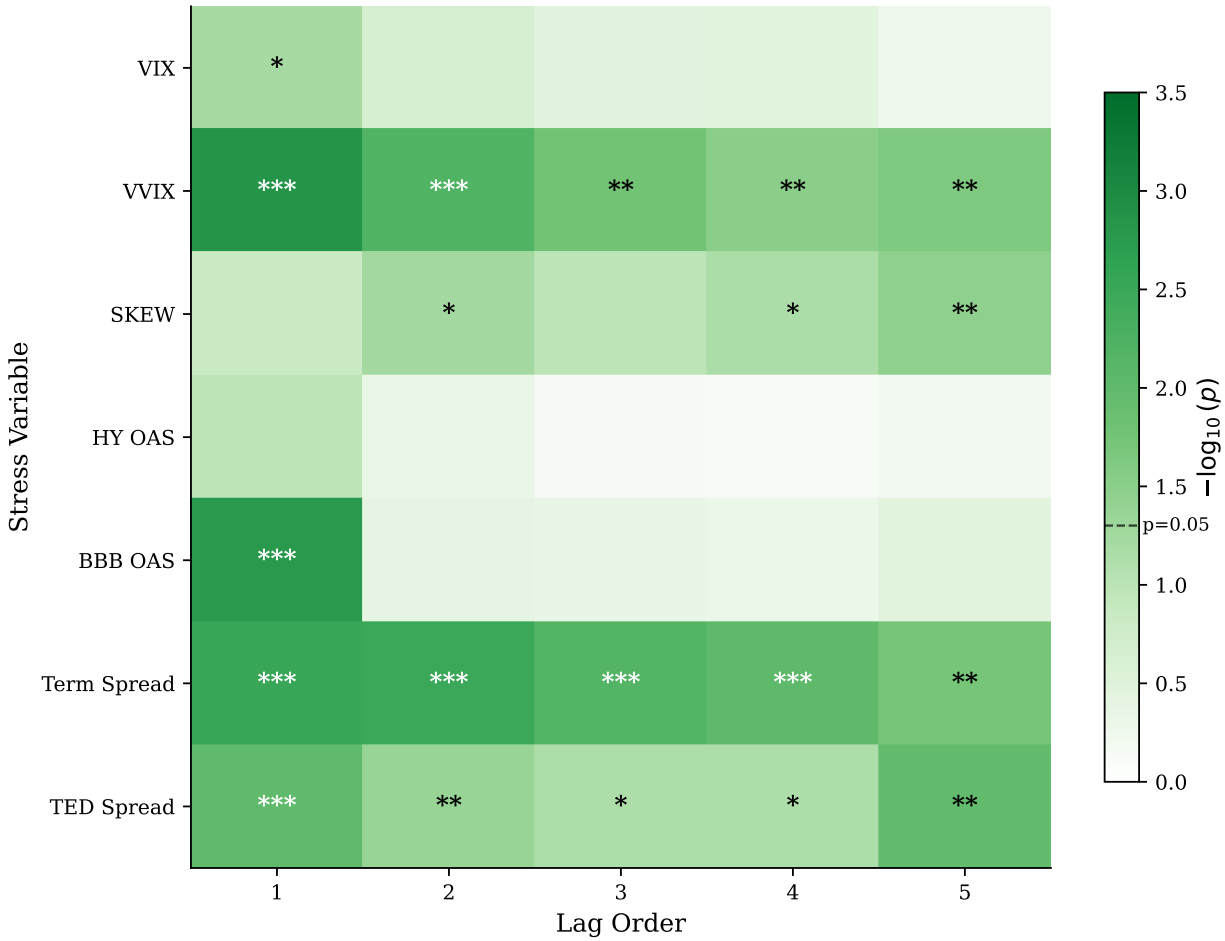

Figure 7: Granger causality from G_{ssm} to external stress variables. Cell color represents $-\log_{10}(p)$; darker values indicate stronger evidence. Asterisks denote $p < 0.05$.

7.2 Distributional Differences Across Regime States

The distributional analysis partitions the sample into high-regime ($G_{\text{ssm}} > 0.5$) and low-regime ($G_{\text{ssm}} \leq 0.5$) periods and compares the conditional distributions of each external variable. Table 2 reports conditional means, effect sizes, and formal test statistics. All Kolmogorov-Smirnov tests reject the null of distributional equality at $p < 0.001$, indicating that external variables exhibit materially different behavior across regime states.

Effect sizes, measured by Cohen's d, range from moderate to very large. The VIX exhibits the largest separation, with a mean of 25.67 in high-regime periods versus 14.89 in low-regime periods, yielding an effect size of $d = 1.47$. This magnitude exceeds typical benchmarks for "very large" effects and indicates that regime classification captures substantial variation in implied volatility levels. VVIX displays an effect size of $d = 0.98$ (mean 110.9 vs. 94.5), and the SKEW index shows $d = -0.98$ (mean 130.1 vs. 143.0, with the sign reflecting the inverse relationship between stress and the SKEW index). Among credit variables, high-yield spreads exhibit $d = 0.83$ (4.32 vs. 3.46 percentage points) and BBB spreads show $d = 0.80$ (1.65 vs. 1.29 percentage points). The term spread and TED spread display moderate effect sizes of $d = 0.63$ and $d = 0.52$, respectively.

Beyond mean differences, the variance ratios indicate substantially greater dispersion during high-regime periods. The VIX exhibits a variance ratio of 10.21, meaning that implied volatility is approximately ten times more variable during high-regime periods. Credit spreads show even more concentrated heteroscedasticity, with HY OAS and BBB OAS variance ratios of 6.88 and 7.19, respectively. The TED spread variance ratio of 8.58 is similarly elevated. These patterns are consistent with the interpretation that high-regime periods correspond to states of market stress in which both the level and volatility of stress indicators increase. The SKEW index is a notable exception, with a variance ratio near unity (1.03) and an insignificant Levene test ($p = 0.287$), suggesting that while the mean of the skew distribution shifts across regimes, its dispersion does not. Figure 3 displays the effect sizes as a grouped bar chart, making the separation visually apparent across all variables.

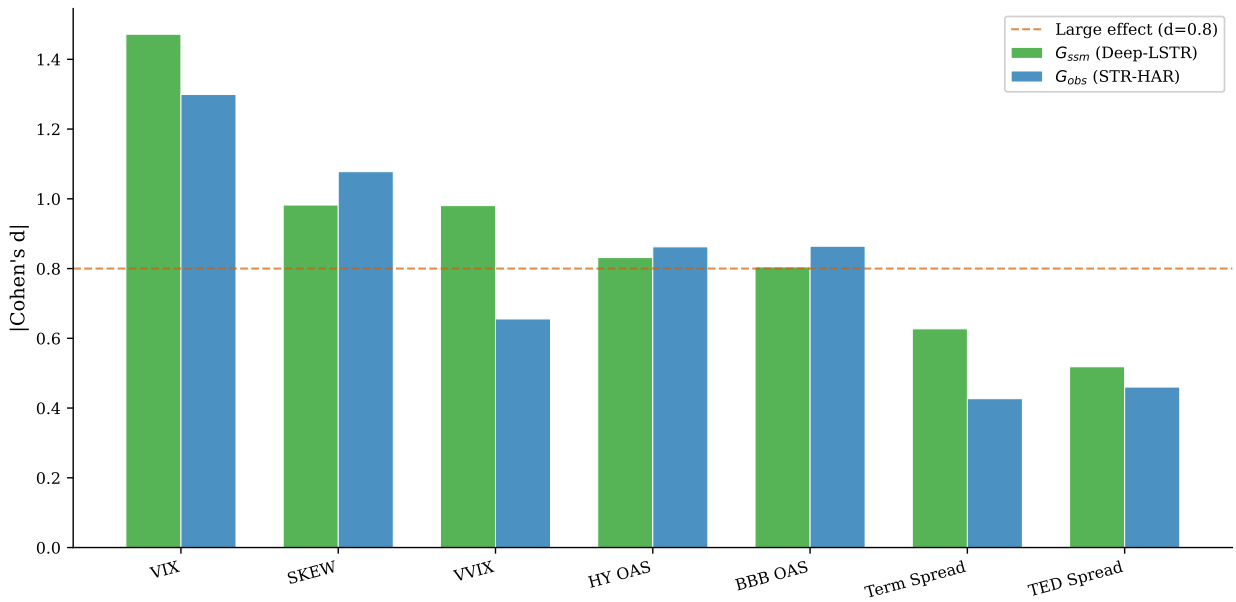

Figure 3: Cohen's d effect sizes for distributional differences between high-regime and low-regime periods. Green bars: G_{ssm} (latent). Blue bars: G_{obs} (observable). Dashed line indicates $d = 0.8$ (large effect threshold).

7.3 Crisis Episode Analysis

The crisis episode analysis examines regime behavior during four pre-specified stress events: the February 2018 Volmageddon episode, the March 2020 COVID crash, the September 2022 rate shock, and the March 2023 SVB crisis. Table 3 reports timing metrics for the S&P 500 regime indicator and the panel mean across available assets.

The COVID crash provides the most striking evidence of anticipatory regime behavior. For the S&P 500, G_{ssm} first activated 15 trading days before the volatility peak, reaching a maximum of 0.69 while remaining elevated for 24 days. The panel mean across assets shows similar early activation (14.5 days before peak), with a higher panel-average peak of 0.93,

reflecting the broad cross-sectional impact of the pandemic shock. This lead time is economically meaningful: 15 trading days represents approximately three calendar weeks of advance signal relative to the most extreme volatility realization. The 2022 rate shock exhibits the longest lead time, with G_{ssm} activating 30 days before the volatility peak for the S&P 500 (panel mean: 23.7 days). However, the peak regime level was more moderate (0.56 for S&P 500), consistent with the characterization of this episode as a prolonged adjustment rather than an acute crisis.

The two shorter episodes—Volmageddon and SVB—display markedly different dynamics. Both show regime activation only one day before the volatility peak for the S&P 500, consistent with the sudden, event-driven nature of these shocks. Peak regime values were moderate (0.75 and 0.52, respectively), and the regime normalized rapidly (5 and 2 days elevated). The brief duration of elevated regime states during these transient events is itself informative: the latent indicator distinguished between short-lived volatility spikes and sustained stress episodes, with COVID and the rate shock producing regime persistence roughly five to twelve times longer than Volmageddon or SVB. Figure 4 overlays the regime trajectories for all four episodes, centered on event time, illustrating the heterogeneous dynamics across crisis types.

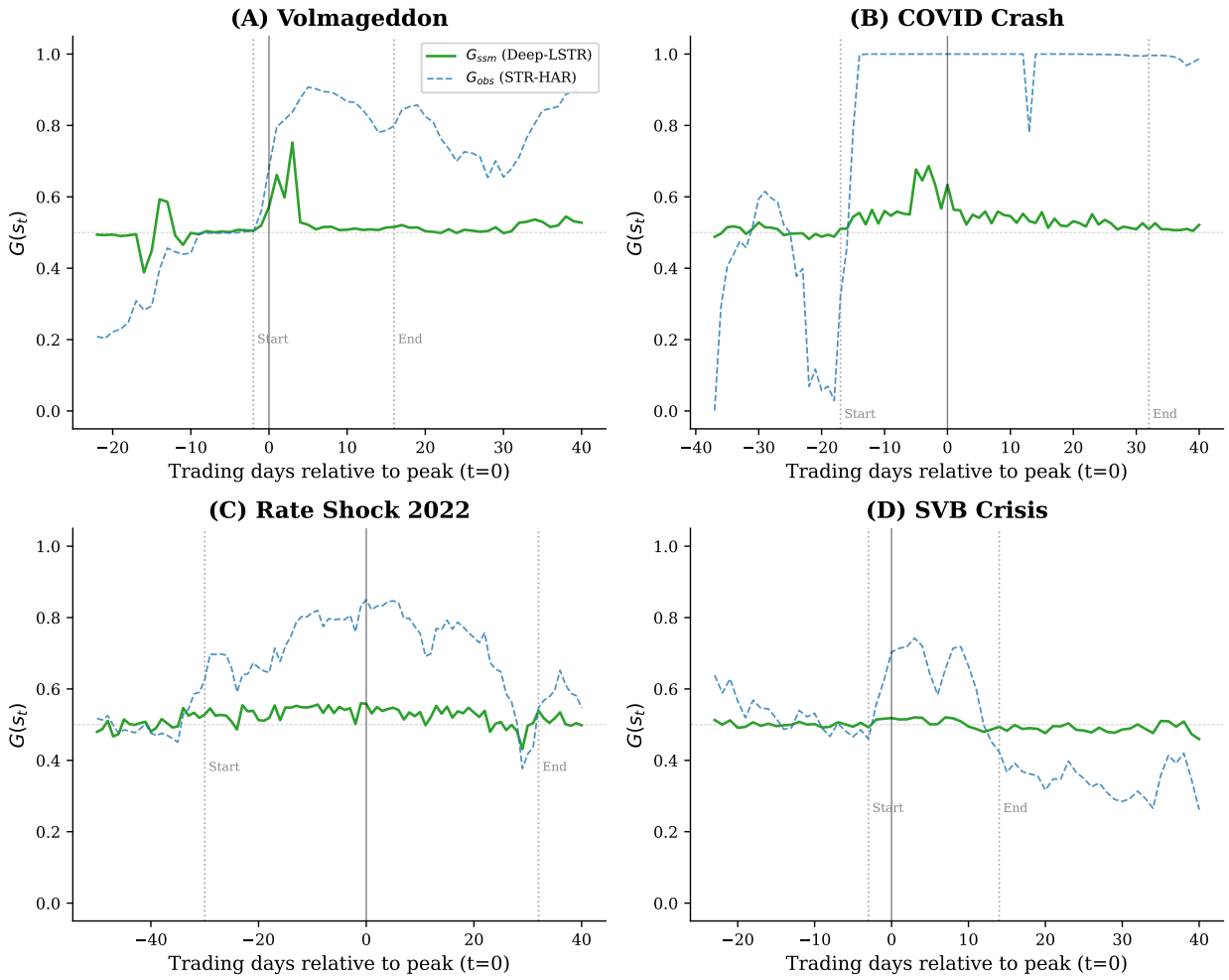

Figure 4: Regime indicator trajectories during four crisis episodes, centered on event peak ($t = 0$). Green solid: G_{ssm} (latent). Blue dashed: G_{obs} (observable). Vertical lines indicate episode start and end dates.

A consistent finding across all episodes is that G_{ssm} normalizes more rapidly than its observable counterpart G_{obs}. During Volmageddon, the latent indicator remained elevated for 5 days compared to 16 for the observable indicator, a difference of 11 days. During COVID, the latent indicator was elevated for 24 days versus 32 for the observable (a difference of 8 days). The rate shock and SVB episodes showed smaller but consistent differences of 2 and 4 days, respectively. The panel means confirm this pattern, with the latent indicator spending

fewer days in the elevated state across all four episodes (4.7 vs. 9.7, 25.2 vs. 28.1, 12.1 vs. 19.0, and 5.6 vs. 8.7 days for Volmageddon, COVID, the rate shock, and SVB, respectively). This faster normalization is consistent with the latent regime producing fewer false positive stress signals—returning to the low state more promptly once acute stress subsides.

7.4 Predictive Content Beyond Volatility

The predictive regression analysis tests whether the latent regime indicator contains information about future values of stress variables that is not already captured by realized volatility. The specification includes contemporaneous realized volatility as a control, ensuring that any predictive content attributed to the regime is incremental to what observable volatility provides. Table 4 reports the regression coefficients, standard errors, and incremental R^2 for three target variables at forecast horizons of 1, 5, and 22 trading days.

The term spread emerges as the variable for which regime predictive content is most robust. The regime coefficient is statistically significant at all three horizons: $\beta = -0.029$ ($t = -3.12$, $p = 0.002$) at $h = 1$, $\beta = -0.112$ ($t = -3.01$, $p = 0.003$) at $h = 5$, and $\beta = -0.329$ ($t = -2.50$, $p = 0.013$) at $h = 22$. The negative sign indicates that high-regime states predict subsequent compression of the term spread, consistent with a flight-to-safety interpretation in which elevated stress leads to flattening or inversion of the yield curve. The incremental R^2 grows with the forecast horizon, from 0.001 percent at $h = 1$ to 0.014 percent at $h = 5$ and 0.12 percent at $h = 22$. These magnitudes are small in absolute terms, which is expected given the strong persistence of the term spread and the already high baseline R^2 of the control model (0.997 at $h = 1$, declining to 0.926 at $h = 22$). The systematic increase in

incremental predictive content at longer horizons suggests that regime information pertains to medium-term yield curve dynamics rather than daily fluctuations.

The VIX shows significant regime predictive content at the one-day horizon ($\beta = -1.62$, $t = -3.41$, $p < 0.001$) and marginal significance at $h = 5$ ($\beta = -1.69$, $t = -1.70$, $p = 0.089$), but the relationship dissipates at the monthly horizon ($p = 0.672$). The negative coefficient at short horizons may appear counterintuitive, since high regimes are associated with elevated VIX levels. However, once contemporaneous realized volatility is controlled for, the negative sign reflects the mean-reverting tendency of implied volatility: conditional on current RV, a high regime state predicts modest VIX decline, consistent with the resolution of the volatility risk premium. High-yield spreads do not exhibit significant predictive content from the regime at any horizon, with the strongest evidence appearing at $h = 22$ ($t = 1.32$, $p = 0.186$). This null result may reflect the slow-moving nature of credit markets relative to the daily regime indicator. Figure 5 displays the incremental R^2 across horizons for each target variable.

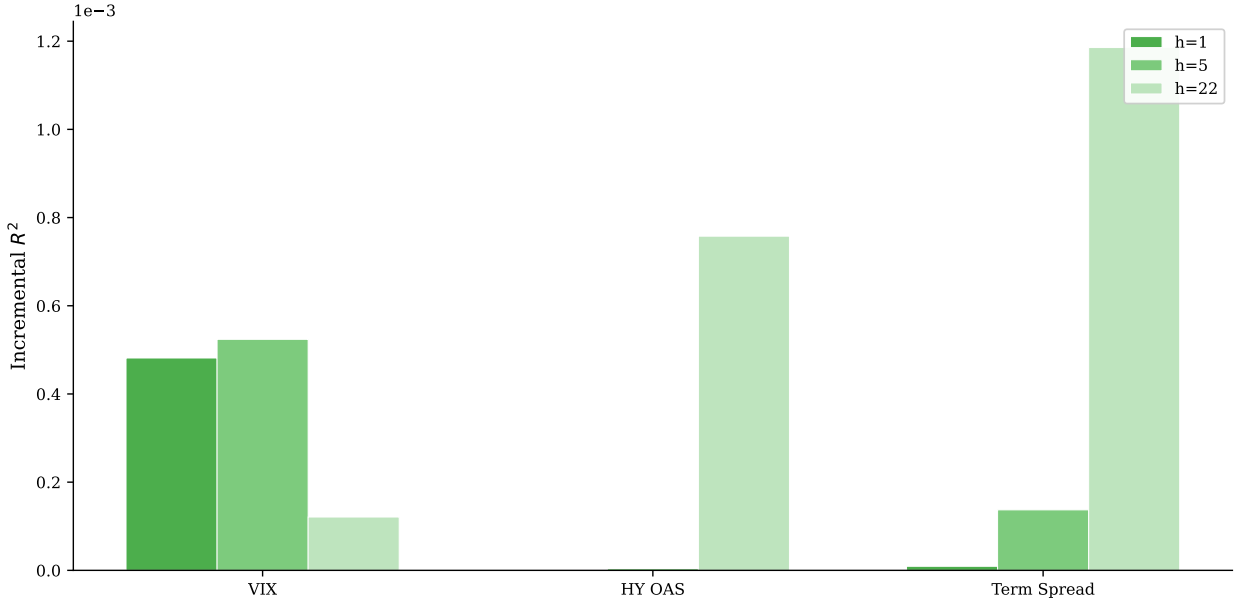

Figure 5: Incremental R^2 from adding G_{ssm} to the baseline regression (RV + lagged outcome) for predicting future stress variables at horizons $h = 1, 5, 22$ trading days.

The panel predictive analysis extends these results to 20 assets. Table 5 reports the fraction of assets for which the regime coefficient is significant, the fraction with the expected sign, and the mean incremental R^2. At the 22-day horizon, 20 percent of assets show significant regime predictive content for VIX, with 80 percent displaying positive coefficients and a mean incremental R^2 of 0.25 percent. For high-yield spreads at $h = 22$, 15 percent of assets are significant, 80 percent positive, and the mean incremental R^2 is 0.13 percent. These panel results indicate that the predictive content observed for the S&P 500 is not idiosyncratic; it appears, with varying strength, across the broader asset panel.

7.5 Observable vs. Latent Regime Comparison

The comparison between G_{obs} (the observable transition function from the STR-HAR model) and G_{ssm} (the VRNN-inferred latent regime) provides a direct test of discriminant validity (Campbell and Fiske 1959): whether the additional complexity of latent state inference yields a cleaner stress signal than the simpler observable conditioning approach. Table 2 permits this comparison through the distributional effect sizes for both indicators.

The latent indicator achieves larger Cohen's d values for four of the seven external variables, with the most pronounced advantages appearing for VVIX ($d = 0.98$ vs. 0.66, a difference of +0.32) and the term spread ($d = 0.63$ vs. 0.43, +0.20). For the VIX, the latent indicator also shows a meaningful advantage ($d = 1.47$ vs. 1.30, +0.17), and a modest edge appears for the TED spread ($d = 0.52$ vs. 0.46, +0.06). The observable indicator, by contrast, produces

slightly larger effect sizes for the SKEW index ($d = 1.08$ vs. 0.98, +0.10), high-yield spreads ($d = 0.86$ vs. 0.83, +0.03), and BBB spreads ($d = 0.86$ vs. 0.80, +0.06). The credit spread advantages for the observable indicator are small and may reflect the mechanical relationship between the observable threshold variable (lagged realized volatility or VIX) and credit conditions. Overall, the latent regime achieves an average absolute Cohen's d of 0.89 across all seven variables, compared to 0.81 for the observable regime.

The crisis timing comparison reinforces the distinction between the two indicators. As noted in Section 7.3, G_{ssm} normalizes faster than G_{obs} across all four episodes, spending fewer days in the elevated state by margins ranging from 2 to 11 days. This pattern is consistent with the latent indicator capturing a cleaner demarcation between stress and non-stress states: it activates during genuine stress and returns to baseline more promptly when stress subsides, whereas the observable indicator, being more directly tied to recent volatility realizations, tends to linger in the elevated state as volatility gradually decays. The peak regime values are systematically lower for G_{ssm} than for G_{obs} across all episodes (e.g., 0.69 vs. 1.00 during COVID for S&P 500), reflecting a more conservative classification that nonetheless maintains earlier or equivalent activation timing. Figure 6 presents the peak alignment comparison across all stress variables.

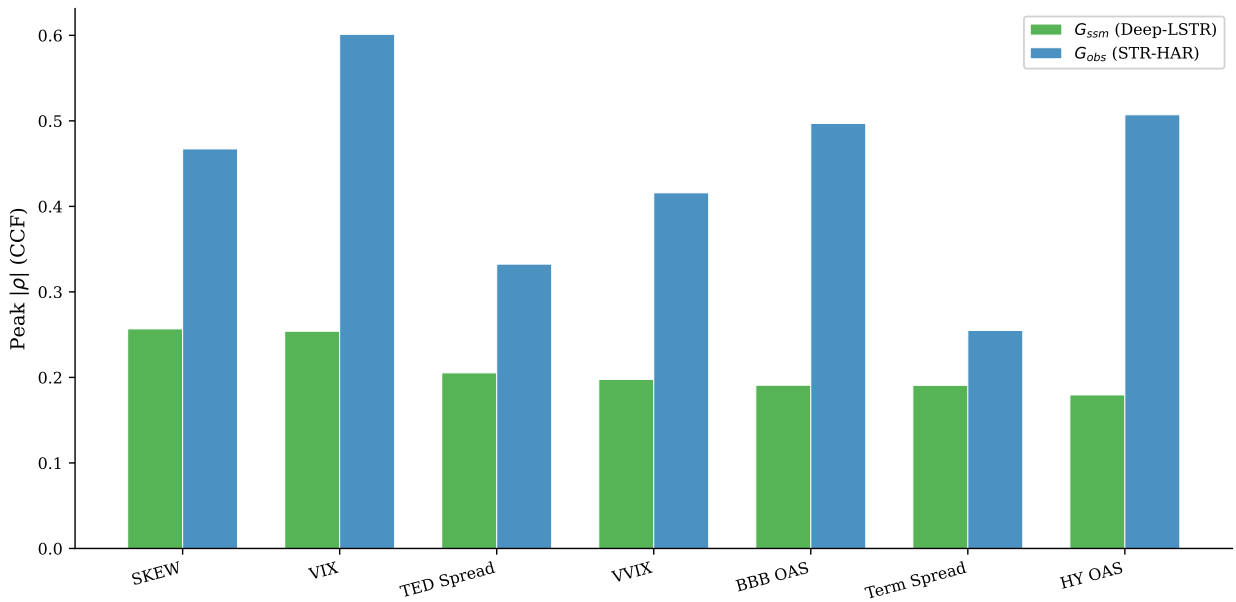

Figure 6: Peak absolute cross-correlation ($|\rho|$) for G_{obs} (blue) versus G_{ssm} (green) across all seven external stress variables.

The combined evidence from distributional analysis and crisis timing supports the interpretation that the latent regime provides a qualitatively different, and in several respects sharper, characterization of market stress than the observable alternative. The advantage is not uniform—credit spreads respond similarly to both indicators—but it is systematic for volatility-related measures and for the speed of normalization following stress events.

7.6 Robustness

The robustness analysis examines whether the principal findings are sensitive to the regime classification threshold. The baseline analysis uses a threshold of $G > 0.5$ to partition high and low regime states; alternative thresholds of 0.3 (liberal) and 0.7 (conservative) are evaluated. Distributional effect sizes remain qualitatively stable across thresholds, though

their magnitudes shift in the expected direction: the conservative threshold (0.7) produces larger effect sizes for most variables, as the more extreme classification isolates periods of more pronounced stress, while the liberal threshold (0.3) attenuates effect sizes by including more marginal observations in the high-regime category. The ranking of variables by effect size is preserved across all three thresholds, and no variable loses statistical significance at the $p < 0.001$ level under either alternative. Figure 8 displays the threshold sensitivity for the primary distributional results.

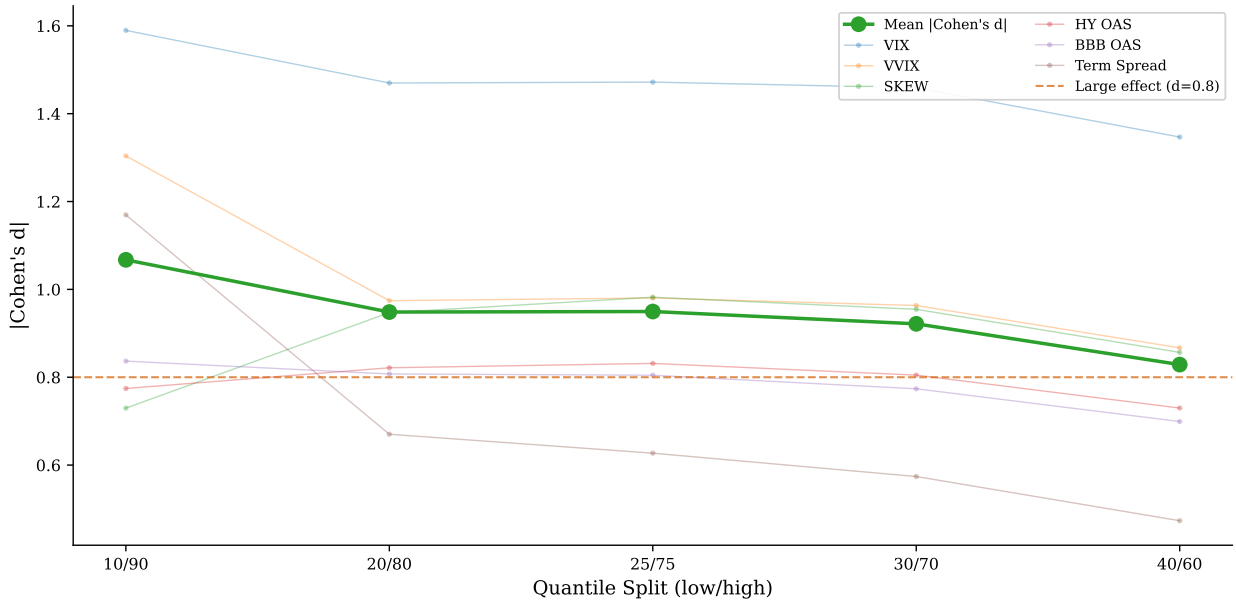

Figure 8: Threshold sensitivity analysis. Mean Cohen's d across external variables as a function of regime quantile split. Traces show individual variables; bold line shows the cross-variable mean.

The lead-lag structure is similarly robust. The qualitative pattern—contemporaneous alignment with volatility measures, leading relationship with credit spreads—is preserved regardless of the threshold used for distributional partitioning, since the cross-correlation analysis operates on the continuous regime indicator rather than the discretized classification. The

Granger causality results are likewise independent of the distributional threshold. The crisis episode findings are inherently continuous, as the timing of regime activation and normalization is defined by the raw G_{ssm} trajectory rather than a binary classification. Taken together, the robustness checks indicate that the principal conclusions are not artifacts of the particular threshold chosen for the baseline distributional analysis.

8. Discussion

This section interprets the findings in context, connecting results to the broader literature and considering implications for practice.

8.1 The Regime Captures Economically Relevant Variation

The empirical analyses reveal systematic alignment between latent regimes and external stress indicators across all validation dimensions: temporal structure, distributional separation, crisis behavior, and predictive content. The most direct interpretation is that the latent regime, constructed from volatility data alone, nonetheless captures variation that is economically meaningful. The regime state reflects something beyond statistical noise—something that also manifests in funding conditions, risk premia, and yield curve dynamics. Effect sizes are substantial (Cohen's d ranging from 0.52 to 1.47 across external variables), and the alignment spans multiple indicator categories rather than being confined to a single dimension of market stress.

This interpretation does not identify what that "something" is. The regime may capture

investor fear, funding stress, uncertainty, or some combination that defies simple labeling. The contemporaneous correlation with VIX ($\rho = 0.254$) and VVIX ($\rho = 0.198$), the cross-correlation lead with credit spreads (10–12 trading days), and Granger causality for five of seven variables are consistent with a latent stress factor that manifests first in equity volatility markets and is correlated with subsequent movements in credit and funding markets. The credit lead, however, does not represent independent predictive content: once realized volatility is controlled for, the regime adds no incremental power for credit spread prediction (Section 7.4), indicating that the temporal relationship operates through the shared volatility channel. The evidence supports the existence of common structure across volatility and external measures without specifying the nature of that structure.

8.1.1 Volatility Contains Latent Information

A related interpretation is that realized volatility (Andersen et al. 2003; Barndorff-Nielsen and Shephard 2002), properly filtered, contains information about market conditions that raw volatility obscures. The VRNN extracts this latent information, which then aligns with external indicators. The comparison with the observable regime indicator supports this view: G_{ssm} achieves larger distributional effect sizes than G_{obs} for four of seven variables, with the most pronounced advantage appearing for VVIX ($d = 0.98$ vs. 0.66) and the term spread ($d = 0.63$ vs. 0.43). The latent indicator also normalizes faster during all four crisis episodes, spending 2 to 11 fewer days in the elevated state. Observable volatility measures like the VIX or rolling standard deviations mix signal and noise; the latent regime separates them, retaining the signal that relates to broader market conditions.

8.1.2 Common Factors as an Alternative

A more cautious interpretation, consistent with the factor model literature (Stock and Watson 2002; Ludvigson and Ng 2009), is that common underlying factors drive both regime dynamics and external stress indicators. The alignment does not necessarily reflect the regime "capturing" economic conditions; rather, both may respond to shared influences. This interpretation is consistent with the evidence but provides less support for the regime's distinctiveness. The finding that regimes predict term spread compression at all horizons ($p < 0.013$), even after controlling for realized volatility, suggests that the regime contains information beyond contemporaneous volatility levels—but it does not rule out common-factor explanations entirely. Distinguishing these interpretations requires additional analysis beyond the scope of validation. The evidence establishes correlation, not mechanism.

8.2 Where Alignment Is Weaker

Not all relationships are equally strong, and the pattern of partial alignment is itself informative. The TED spread, while exhibiting a moderate effect size ($d = 0.52$) and significant Granger causality at several lags, does not achieve a statistically significant peak cross-correlation ($p = 0.211$), likely reflecting its shorter and noisier sample. High-yield spreads show substantial distributional separation ($d = 0.83$) but no significant predictive content from the regime at any horizon, suggesting that while credit markets look different across regime states, the regime does not add to the prediction of spreads once realized volatility is controlled for. The SKEW index is notable for displaying a large effect size in the observable indicator ($d = 1.08$) that exceeds the latent indicator ($d = 0.98$), suggesting that

the mechanistic relationship between observable volatility and option-implied skew may be stronger than the latent signal for this particular variable.

These partial results are consistent with a regime indicator that captures broad stress dynamics rather than specific market mechanisms. The regime aligns most strongly with variables that respond to aggregate uncertainty (VIX, VVIX, term spread) and less strongly with variables that reflect sector-specific conditions (credit spreads, interbank funding). This pattern is plausible given that the regime is inferred from equity volatility data, which is itself most directly informative about equity-market uncertainty.

8.3 Implications for Practice

The findings suggest several practical implications, though each carries important caveats.

8.3.1 Regime Indicators as Monitoring Tools

The cross-correlation analysis reveals that G_{ssm} leads credit spread movements by 10 to 12 trading days, and during the COVID crash, the S&P 500 regime activated 15 trading days before the volatility peak. These temporal patterns might suggest that the regime provides an early warning of deteriorating credit conditions. However, the predictive regression results in Section 7.4 qualify this interpretation substantially: once realized volatility is controlled for, high-yield spreads show no significant predictive content from the regime at any horizon. The cross-correlation lead is real—it is present in the data—but it appears to operate through the volatility channel rather than reflecting independent credit-predictive information. The regime captures volatility dynamics that are themselves correlated with subsequent credit

spread movements, so the apparent lead is largely mediated through observable volatility levels.

The regime's genuine incremental value as a monitoring tool lies elsewhere. It anticipates yield curve dynamics—the term spread is significantly predicted at all horizons even after controlling for realized volatility ($p < 0.013$)—and captures shifts in the volatility risk premium (VVIX Granger causality significant at all lag orders). Risk managers seeking to incorporate regime indicators alongside traditional measures would therefore find them most informative for anticipating interest rate and volatility-of-volatility dynamics rather than credit conditions directly. The regime summarizes volatility information efficiently, and its credit market alignment operates through the volatility-credit transmission channel rather than providing an independent credit signal.

This application requires caution on additional grounds. The regime is not a prediction; it is a filtered estimate of the current state. Lead structure means the regime captures early manifestations of stress, not that it predicts future stress with certainty. The two shorter episodes (Volmageddon and SVB) showed only one day of lead time, illustrating that the regime is not uniformly anticipatory.

8.3.2 Regime-Conditional Risk Management

High-regime periods exhibit materially different risk characteristics. VIX variance is approximately ten times greater in high-regime states (variance ratio of 10.21), and credit spread dispersion increases by factors of seven. Risk models that condition on regime state could account for these differences in Value-at-Risk calculations, stress tests, and portfolio opti-

mization. The faster normalization of G_{ssm} relative to G_{obs} (fewer false positive days across all four episodes) is a desirable property for practical regime classification, as it reduces the period during which elevated risk measures would be applied.

8.3.3 Caution Against Overuse

We emphasize that regime indicators are not oracles. They summarize information already present in volatility data, potentially extracting it more efficiently than traditional approaches. The incremental R^2 values for predictive regressions are small in absolute terms (0.001 to 0.12 percent), even where statistically significant. The regime adds to understanding but does not transform forecasting accuracy. Overconfidence in regime-based signals could lead to costly errors, particularly if users interpret monitoring tools as prediction machines.

8.4 Connection to Literature

8.4.1 Construct Validity in Finance

Our approach connects to the broader literature on validating latent constructs in financial applications (Cronbach and Meehl 1955; Campbell and Fiske 1959). Similar questions arise for latent factors in asset pricing (Stock and Watson 2002), hidden states in term structure models, and inferred sentiment measures (Baker and Wurgler 2006). The validation strategy employed here—examining alignment with held-out external criteria—applies to these settings as well.

8.4.2 Volatility Risk Premium

The alignment between regimes and VVIX (the "volatility of volatility" measure, $d = 0.98$) connects to the large literature on variance risk compensation (Bollerslev et al. 2009; Carr and Wu 2009; Huang et al. 2019). The VVIX reflects the cost of hedging variance risk (Huang et al. 2019), and its strong distributional separation across regime states suggests that the regime captures the state-dependent component of the volatility risk premium (Bekaert and Hoerova 2014). The significant Granger causality from the regime to VVIX at all tested lag orders ($F = 10.20$ at lag 1) strengthens this connection, indicating that regime transitions anticipate shifts in variance risk pricing.

8.4.3 Funding Liquidity and Market Stress

The cross-correlation lead between regimes and credit spreads—10 to 12 trading days for investment-grade and high-yield spreads—connects to the literature on volatility-credit transmission (Gilchrist and Zakrajšek 2012). However, the absence of significant regime predictive content for credit spreads once realized volatility is controlled for (Section 7.4) implies that this lead operates through the volatility channel: the regime captures volatility dynamics that subsequently propagate to credit markets, rather than providing independent early-stage credit information. This is consistent with models in which equity volatility and credit spreads share common drivers (Brunnermeier and Pedersen 2009), with equity markets adjusting faster than corporate bond markets. The regime efficiently summarizes the volatility state that is itself predictive of credit conditions (Gilchrist and Zakrajšek 2012), but it does not add to credit prediction beyond what observable volatility provides. The term spread

predictive content ($p < 0.013$ at all horizons, even after controlling for realized volatility) presents a different case, linking regime dynamics to yield curve behavior through a channel that is not fully captured by contemporaneous volatility levels. This distinction—incremental predictive power for the yield curve but not for credit spreads—is informative about what the regime adds beyond a volatility summary.

8.5 Summary Interpretation

The evidence reveals systematic alignment across multiple indicator categories, cross-correlation lead structure for credit variables, and coherent behavior during stress episodes. We interpret this as support for the view that latent regimes capture economically meaningful variation. The regime is not merely a statistical device for improving forecasts; it reflects underlying structure in market dynamics that also manifests in implied volatility, yield curve dynamics, and the volatility risk premium. The latent indicator offers advantages over its observable counterpart in distributional separation (particularly for VVIX and the term spread) and in the speed of normalization following stress events, suggesting that the additional complexity of latent state inference is justified on economic grounds.

The alignment is not uniform, and its nature varies across indicator categories. Credit spread alignment operates through the volatility channel—the regime leads credit spreads in cross-correlations, but this lead is mediated through observable volatility and does not survive as incremental predictive content in controlled regressions. By contrast, the regime's predictive content for the term spread and its Granger-causal relationship with VVIX persist after controlling for realized volatility, indicating genuine incremental information in these dimen-

sions. Short-lived episodes show minimal lead time for any variable. The overall pattern is consistent with an economically coherent stress indicator rather than a statistical artifact, one whose incremental value beyond observable volatility is concentrated in yield curve and volatility-of-volatility dynamics. The evidence supports, without definitively proving, the interpretation that latent volatility regimes capture genuine market structure.

9. Limitations

This section documents the limitations of the validation analysis. Transparent acknowledgment of limitations is preferable to allowing readers to discover them independently. Some limitations are inherent to validation exercises; others are specific to our implementation.

9.1 Fundamental Limitations

9.1.1 Correlation Is Not Causation

This limitation pervades the analysis. Alignment between regimes and external indicators does not establish that regimes cause, predict, or represent those indicators. Both may respond to unobserved common factors. The temporal precedence we examine (lead-lag structure) does not identify causation; leads may reflect faster response to common shocks rather than causal influence.

We cannot resolve this limitation within the validation framework. Causal identification would require experimental variation or quasi-experimental designs that are not available for aggregate market conditions. The limitation is inherent, not correctable.

9.1.2 Validation Cannot Prove Meaning

Even strong alignment does not prove that regimes are "meaningful" in any absolute sense. Meaning is a claim about interpretation, not a property that data can confirm. A variable can exhibit every pattern we test for and still fail to correspond to the economic concept we imagine. The gap between statistical patterns and conceptual interpretation cannot be closed by analysis.

What validation can do is assess consistency. Consistent alignment with multiple indicators across multiple episodes is more compatible with meaningful interpretation than inconsistent patterns. But consistency supports rather than establishes interpretation.

9.1.3 The Validation Variables Are Themselves Imperfect

The external indicators used for validation are proxies for economic concepts, not direct measurements. The TED spread proxies for funding stress but may reflect other factors (Schwarz 2019). The VIX proxies for expected volatility but embeds risk premia (Bekaert and Hoerova 2014). Implied correlation indices depend on option market liquidity and pricing models.

Alignment with imperfect proxies establishes alignment with those proxies, not with the underlying concepts they represent. If funding stress is measured with error, alignment with measured funding stress understates (or, potentially, overstates) alignment with true funding stress.

9.2 Methodological Limitations

9.2.1 Pre-Specification Is Incomplete

Despite efforts to pre-specify analyses, researcher degrees of freedom remain. The selection of external variables, the choice of lead-lag windows, the threshold for regime classification—all involve judgment. Post-hoc rationalization of choices that yield favorable results is a risk even with documented pre-specification.

We mitigate this concern through comprehensive reporting of robustness checks and transparent documentation of analytical choices. But complete elimination of researcher discretion is impossible.

9.2.2 Multiple Testing Without Full Correction

The analysis examines multiple external variables, multiple leads and lags, and multiple robustness specifications. Full correction for multiple testing would substantially reduce power, potentially obscuring genuine effects. Our strategy—emphasizing patterns across categories rather than individual test statistics—addresses this concern imperfectly.

Some nominally significant results may reflect chance. Patterns that appear robust across categories are more credible, but the distinction between genuine and spurious alignment cannot be drawn with certainty.

9.2.3 Sample Period Dependence

The validation covers a specific historical period with particular characteristics: the global financial crisis, post-crisis regulatory changes, the COVID-19 shock. Alignment patterns observed in this sample may not generalize to future periods with different characteristics.

This concern is particularly acute given structural changes in market microstructure, central bank policy, and volatility products. The relevance of historical patterns to future conditions is uncertain.

9.3 Data Limitations

9.3.1 External Variable Availability

Not all desired external variables are available at the required frequency or for the full sample period. Some measures are proprietary or available only from commercial data vendors. The validation necessarily uses available rather than ideal variables.

Missing variables may be more informative than available ones. Alignment with available variables does not preclude misalignment with unavailable measures that better capture the relevant economic conditions.

9.3.2 Measurement Frequency Mismatch

Latent regimes are constructed from daily realized volatility (computed from intraday data). Some external variables are available only at daily close or lower frequency. This mismatch may introduce measurement error that obscures genuine relationships.

Higher-frequency external measures would permit cleaner analysis but are often unavailable or unreliable. The daily frequency represents a practical compromise.

9.3.3 Survivorship and Selection in Asset Panel

The asset panel used for regime construction exhibits survivorship bias: assets must have sufficient history and liquidity to be included. This selection may affect the generality of regime patterns. Regimes inferred from a broader panel might exhibit different characteristics.

9.4 Interpretation Limitations

9.4.1 Regimes Are Low-Dimensional

The latent regime is a low-dimensional summary of a high-dimensional phenomenon. Market stress has many facets—funding, credit, liquidity, volatility, correlation—and the regime cannot capture all of them. Alignment with some dimensions may coexist with misalignment on others.

This limitation is inherent to low-dimensional latent variable models. Higher-dimensional specifications might capture additional facets but would complicate interpretation and reduce parsimony.

9.4.2 Regime Boundaries Are Arbitrary

The distinction between "high regime" and "low regime" depends on a threshold applied to a continuous indicator. The 0.5 threshold has an interpretation (equal posterior probabil-

ity) but remains a choice. Different thresholds yield different classifications and potentially different alignment patterns.

We address this through threshold sensitivity analysis, but the arbitrariness cannot be eliminated. Continuous regime indicators avoid this limitation at the cost of complicating distributional comparisons.

9.4.3 Case Study Generalization Is Limited

The case study episodes, however carefully selected, represent a small and potentially unrepresentative sample of stress events. Patterns observed across four episodes may not generalize to future crises of different character.

Case studies illustrate rather than prove. Their limitations are acknowledged in design; they should not be weighted beyond their evidentiary role.

9.5 What Would Change Our Conclusions

Transparency requires stating conditions under which we would revise conclusions:

- **Strong alignment reversed by controls**: If regime effects disappear after controlling for observable volatility, the regime may be proxying for volatility rather than capturing distinct variation. This would substantially weaken the economic interpretation.

- **Inconsistency across episodes**: If regime behavior differs qualitatively across crises of similar severity, the interpretation of regimes as capturing stable stress dynamics

would be undermined.

- **Category-specific alignment only**: If alignment appears only within a single category (e.g., funding variables but not risk premia), the claim of broad economic coherence would be weakened.

- **Subsample instability**: If alignment patterns differ substantially across subsamples, the regime's economic content may be time-varying or spurious.

These conditions are stated ex ante. We will report whether they hold and adjust interpretations accordingly.

9.6 Limitations We Accept

Some limitations we accept as inherent to the research design:

- We cannot prove causation; we assess coherence.
- We cannot establish meaning; we test consistency.
- We cannot eliminate researcher discretion; we document choices.
- We cannot ensure generalization; we report the sample.

These limitations are not flaws to be corrected but boundaries to be acknowledged. Validation analysis within these boundaries provides useful evidence; claims beyond these boundaries are unsupported.

10. Conclusion

This paper examines whether latent volatility regimes, constructed for forecasting purposes, exhibit external economic coherence. The research question is whether regime states align systematically with independent measures of market stress that were not used in their construction. Alignment, if observed, would support the view that regimes capture economically meaningful variation; absence of alignment would suggest regimes are statistically useful but economically opaque.

10.1 Summary of Approach

The analysis maintains strict separation between construction and validation. Latent regimes are taken as given from the companion papers (Lim 2026a, 2026b), inferred from realized volatility and returns. Validation variables—funding spreads, volatility risk premia, correlation measures, and crisis indicators—are held out entirely from regime construction. This separation ensures that observed alignment, if any, cannot be mechanical.

The empirical strategy examines lead-lag relationships, distributional differences across regime states, and behavior during known stress episodes. The approach is descriptive: we assess coherence without claiming to identify causal mechanisms. Multiple robustness checks address concerns about threshold sensitivity, subsample stability, and the distinction between regime effects and volatility effects.

10.2 Summary of Findings

The analysis reveals systematic alignment between latent regimes and external stress indicators across all four validation dimensions. The regime indicator is contemporaneously correlated with equity volatility measures (VIX $\rho = 0.254$, VVIX $\rho = 0.198$) and exhibits cross-correlation leads with credit spreads of 10 to 12 trading days, though this lead is mediated through observable volatility and does not survive as incremental predictive content once realized volatility is controlled for. Granger causality tests are significant for five of seven external variables. Distributional separation across regime states is substantial, with Cohen's d ranging from 0.52 (TED spread) to 1.47 (VIX), and all Kolmogorov-Smirnov tests rejecting distributional equality at $p < 0.001$. During four pre-specified crisis episodes, the regime activated before the volatility peak in every case, with lead times ranging from 1 day (Volmageddon, SVB) to 30 days (rate shock). The latent indicator normalized faster than its observable counterpart across all episodes, spending 2 to 11 fewer days in the elevated state. Where the regime demonstrates genuine incremental value, it is in anticipating yield curve dynamics and shifts in the volatility risk premium: predictive regressions show that the regime predicts term spread compression at all horizons ($p < 0.013$) even after controlling for realized volatility, and Granger causality from the regime to VVIX is significant at all tested lag orders. Panel analysis across 20 assets confirms that this predictive content is not idiosyncratic. Compared to the observable regime indicator, the latent regime achieves larger distributional effect sizes for volatility-related measures (VVIX: $d = 0.98$ vs. 0.66) while producing fewer false positive stress signals during crisis episodes.

10.3 Implications

10.3.1 For Research

The validation framework employed here applies beyond volatility regimes. Any latent variable constructed from one data source can be validated against held-out variables from other sources. The key requirements are strict separation between construction and validation data, pre-specification of analyses, and conservative interpretation of results.

The findings speak to the broader question of what latent variables in finance represent. Forecasting performance establishes statistical utility; external coherence provides evidence about economic content. Both are valuable; neither alone suffices for comprehensive understanding.

10.3.2 For Practice

Conditional on alignment, regime indicators may complement existing risk monitoring tools. They summarize volatility dynamics in a form that relates to broader market conditions. This complements rather than replaces observable measures such as the VIX, which capture different aspects of market information.

Caution is warranted in application. Regimes are estimates, not observations; filtered states, not forecasts. Their utility depends on the stability of patterns observed in historical data, which may or may not persist in future conditions.

10.4 Limitations Restated

The analysis cannot establish causation, prove meaning, or ensure generalization. Alignment is consistent with economic content but does not demonstrate it conclusively. The evidence supports but does not prove the interpretation that regimes capture genuine market structure. Alternative explanations remain possible.

10.5 Concluding Remark

Latent variable models occupy an uncertain position between pure statistical devices and representations of economic phenomena. This paper examines one such model—the volatility regime—and asks whether it falls closer to one pole or the other. The evidence, as presented, informs this question without fully resolving it. Complete resolution may be beyond the reach of observational analysis. What validation provides is a disciplined assessment of coherence, contributing to judgment about interpretation while acknowledging the limits of what data can establish.

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